

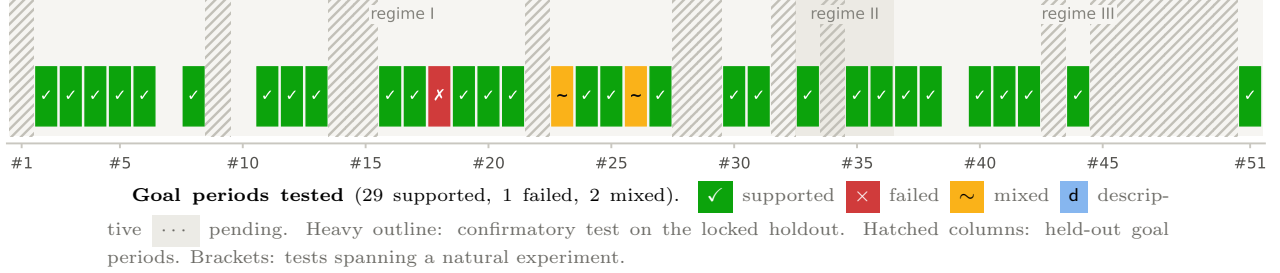
H64: Conflict needs a scarce prize

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. An assigned prize switches antagonism on along opponent bonds and its settlement switches it off by the first reply after the verdict is read; open competition without sides adds none.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The test asks whether stance antagonism between agents needs a rival-exclusive prize and ends when the prize is settled. Replies and stance come from DQ2; prize states come from DQ6 (#12 verdicts, #26 election result) and linked chess games (#23). Dated predictions and a synthetic power study came first. **Debates (#12):** opponents answer each other 0.78 [0.56, 1.10] worse than teammates while the prize is open, and 0.09 [−0.26, 0.39] after the verdict ($p = 0.001$). Teammates do not change, and losers do not resent winners. **Competitions** (#6, #23, #26, #27) have more confident position-opposition than every-day weeks (4/4 above the median, $p = 0.022$). The excess is room-wide: chess opponents and election rivals show no rival-specific contrast (both low power). Without a prize, no antagonism exceeds the agent-field null. **Round 1c** uses the validated v2.1 disagreement flag (precision 0.61–0.67) with the labeller’s confusion built into the null. In #12 it marks 33 of 75 opponent replies and 0 of 60 teammate replies during speeches, and 0 of 64 after the verdict ($\Delta = +0.31$ [0.16, 0.52], $p = 0.0002$). The competition contrast does not survive (2/4 above the median, $p = 0.39$). The verdict moves from supported to **narrowed**: settlement switches assigned conflict off. The holdout is not run.

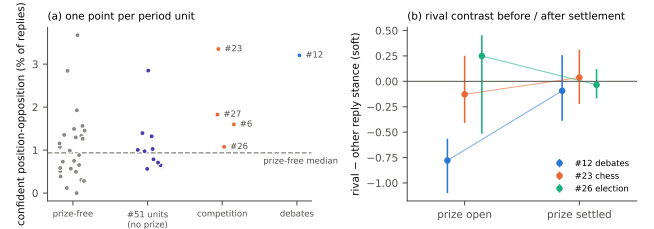
PHYSICAL MODEL

Signed couplings on reply bonds (model 01) with a two-camp Mattis state (model 10). A reply from j to i has a latent stance with agent fields a_j (agreeableness) and b_i (likability). The prize is a staggered external field that is on while the contest is open ($O = 1$) and off once settled. $R_{ij} = 1$ marks rivals for one exclusive outcome. Δ is field-induced antagonism, δ a pair constant that survives settlement (remanence), and ϕ a contest field on every bond (heat).

$$y_e^* = \mu + a_j + b_i + \Delta R_{ij} O(t) + \delta R_{ij} + \phi O(t) + u_{k(e)} + \varepsilon_e$$

$$\gamma_{\text{open}} = \delta + \Delta, \quad \gamma_{\text{set}} = \delta, \quad \hat{\Delta} = \hat{\gamma}_{\text{open}} - \hat{\gamma}_{\text{set}}$$

$$r_p = \Pr [\text{confident opposes, position} \mid \text{reply}]$$



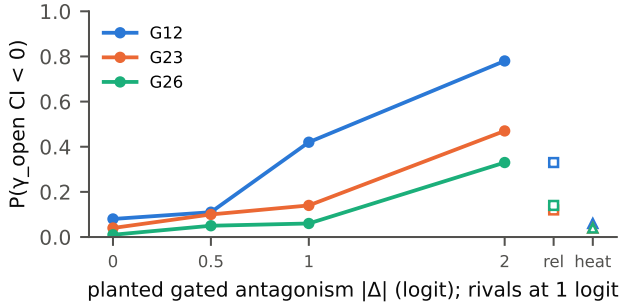
Observables. (a) Confident position-opposition per period unit, by prize class (pre-registered). All four competition periods lie above the prize-free median; #51 rivals without a prize sit inside the prize-free range. (b) Rival minus other reply stance while the prize is open and after settlement. In #12 the contrast falls from −0.78 to −0.09 at the verdict. The #23 and #26 contrasts are small and uncertain.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To end a fight between agents, settle the prize. In #12 the stance contrast between opponents falls to its noise floor within the post-verdict phase (≤ 10 min), and losers show no residual hostility. Agents read the verdict within about 20 s, and no reply after it is flagged. Open contests without assigned sides show no excess disagreement in the validated channel (round 1’s room-wide excess came from an unvalidated subtype). Rival roles without a single winner (#51) create no feuds. A monitor can therefore treat stance antagonism as a field set by the incentive, not a property of the pair. Outside assigned debates the validated flag is rare (about 1% of replies), so small effects stay invisible.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 competition > prize-free r_p	4/4 above median 0.94%; MW $p = 0.022$ (regime I only: 0.010)	yes
P2 no excess pairs without a prize	25/26 units; #51 11/12 (low-power test)	yes
P3 #12 highest r_p	third (#40, #23 higher)	no
P4 #51 at the prize-free level	1.06% < 75th pct 1.35%	yes
N1a #12 open contrast < 0	-0.78 [-1.10, -0.56], $p = 0.0002$	yes
N1b vanishes at verdict	$\Delta = -0.69$ ($p = 0.001$); $\gamma_{\text{set}} = -0.09$, CI $\ni 0$	yes
N1c/d no heat, no resentment	teammates +0.00; losers +0.19 [-0.04, 0.33]	yes
N2a #26 runoff rivals	+0.25 (14 replies), $p = 0.86$	inconcl.
N3a #23 chess opponents	-0.13 (44 replies), $p = 0.31$	inconcl.
1c N1a/N1b (v2.1 flag)	open +0.34 [0.21, 0.51]; $\Delta = +0.31$, $p = 0.0002$; noise-null $p = 0.0015$	yes
1c P1 (v2.1 rate)	2/4 above median 0.96%; $p = 0.39$ (corrected 0.56)	killed
1c P2 / P3 / P4	23/26; #12 second; #51 0.85% < 1.51%	yes / no / yes
1c N1e read timing	0 unread post-verdict replies (lag 20 s)	untestable



Second figure. Synthetic power of the prize-gated DiD on the real reply structures (hard labels, DQ2-like noise, 100 replicates). #12 detects a 2-logit gated antagonism 78% of the time; #23 and #26 at most 47%. Planted remanence (squares) and heat (triangles) do not produce a negative Δ . The card's equivalence clause was unidentifiable and was replaced before the real run.

GOAL PERIODS IN DETAIL

G02 ✓ supported
 G03 ✓ supported
 G04 ✓ supported
 G05 ✓ supported
 G06 ✓ supported
 G08 ✓ supported
 G11 ✓ supported
 G12 ✓ supported
 G13 ✓ supported
 G16 ✓ supported

+22 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

Round 1's competition contrast used DQ2's position subtype (κ 0.11); the validated flag kills it. The v2.1 flag has recall about 0.6; at 0.05 a team-permutation test is anti-conservative under the labeller's confusion (size 0.125), so the #12 result is also checked against the label-noise null. All four competition periods are regime I; topic (scores, rules) could raise disagreement without an incentive. The cluster-robust pair test misses even #12, so the prize-free floor is weak evidence. #26 and #23 have 14 and 44 rival replies while open. The #12 switch-off was known from H21 and H37; the new parts are the heat and remanence tests and the competition contrast.

ROUND 2

- **Essence.** what in an incentive structure makes LLM agents disagree, and how fast the disagreement relaxes when the incentive is removed.
- **H64-R1** Validate the conflict sensor.
- **H64-R2** Rival-specific vs room-wide.
- **H64-R3** Relaxation time.
- **H64-R4** Run `confirm.py` after committing (#29, #34, #51 tail).

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